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# Microbial communities associated with methylmercury degradation in paddy soils

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## ABSTRACT

Bioaccumulation of the neurotoxin methylmercury (MeHg) in rice has raised worldwide concerns due to its risks to human health. Certain microorganisms are able to degrade MeHg in pure cultures, but the roles and diversities of the microbial communities in MeHg degradation in rice paddy soils are unknown. Using a series of microcosms, we investigated MeHg degradation in paddy soils from Hunan, Guizhou, and Hubei provinces, representing three major rice production regions in China, and further characterized one of the soils from Hunan province for microbial communities associated with MeHg degradation. Microbial demethylation was observed in all three soils, demonstrated by significantly more MeHg degraded in the unsterilized soils than the sterilized controls. More demethylation occurred in water-saturated soils than unsaturated soils, but the addition of molybdate and bromoethanesulfonic acid as the respective inhibitors of sulfate reducing bacteria and methanogens showed insignificant effects on MeHg degradation. However, the addition of Cu enhanced MeHg degradation and the enrichment of *Xanthomonadaceae* in the unsaturated soil. 16S rRNA Illumina sequencing and metatranscriptomic analyses of the Hunan soil consistently revealed that *Catenulisporaceae, Frankiaceae, Mycobacteriaceae,* and *Thermomonosporaceae* were among the most likely microbial taxa in influencing MeHg degradation in the paddy soil, and they were confirmed by combined analyses of the co-occurrence network, random forest modeling, and linear discriminant analysis of effect size. Our results shed additional light onto the roles of microbial communities in MeHg degradation in paddy soils and its subsequent bioaccumulation in rice grains.

## INTRODUCTION

Methylmercury (MeHg) is a neurotoxin that can bioaccumulate in rice grain and subsequently threatens human health through rice ingestion, especially for people who consume rice as a staple food.[1] It is thus crucial to understand the pathways and biogeochemical controls on the production and accumulation of MeHg in rice paddy soils. Extensive studies have been devoted to deciphering the mechanisms of microbial Hg methylation driven by the microbial guilds containing the *hgcAB* gene pair, such as sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB) and methanogens.[2-4] However, net MeHg production is determined not only by Hg methylation but also MeHg degradation,[5] which potentially offers a strategy to degrade or minimize MeHg levels in soil and water.

Paddy soil is a typically anoxic habitat where microbial demethylation could be an important pathway of MeHg degradation as observed in other anoxic environments,[6-8] but it remains unknown which microbial communities are responsible and what mechanisms are involved. Previous studies revealed that microbes could degrade MeHg through two major pathways, namely the oxidative demethylation and reductive demethylation.[6, 9] Oxidative demethylation usually occurs in anoxic environments, where MeHg is degraded to Hg(II) and $CO_2$. Studies have shown that SRB and methanogens could be the major drivers of oxidative demethylation in saturated soils.[5, 10] However, recent studies have discovered that aerobic methanotroph (i.e. *Methylosinus trichosporium* OB3b) can also degrade MeHg, probably by an oxidative demethylation pathway, in which MeHg degradation is linked to Cu metabolism.[11, 12] Reductive demethylation is performed by microorganisms containing the Hg resistance genes, in which organomercurial lyase MerB degrade MeHg to Hg(II), which is subsequently reduced to element Hg(0) by the reductase MerA.[6, 13-15] The identified MerA homologs are prevalent among diverse members of *Actinobacteria*, *Firmicutes*, *Proteobacteria*, *Bacteroidetes*, *Chloroflexi*, *Deinococcus/Thermus*, *Tenericutes*, *Nitrospira*, and *Verrucomicrobia*.[16] Reductive demethylation usually occurs in oxic environments at relatively high Hg concentrations.[14] However, a recent culture study reported that an iron-reducing bacterium *Geobacter bemidjiensis* Bem has the ability not only to methylate

Hg but also degrade MeHg via reductive demethylation.[7] While these findings have significantly advanced our understanding on microbial degradation of MeHg, currently little is known about the microbial taxa associated with MeHg degradation in paddy soils. The lack of specific marker genes has made it difficult to directly identify the microbial communities responsible for MeHg degradation *in situ*. It is thus critically needed to examine and understand the shifts of microbial communities during demethylation, which may help to establish a linkage between soil microbial communities and MeHg degradation in paddy soils.

Microbial degradation of MeHg in soil could also be affected by physical and chemical factors that have been suggested to be key regulators of soil microbial communities.[17, 18] In particular, water and nutrient conditions are among the most important factors regulating the diversity and activities of microbial communities.[19, 20] Meanwhile, paddy soils are more influenced by human activities (e.g., artificial drainage and fertilizer application) than other natural soils such as bogs and wetlands. Previous studies demonstrated that water saturation and nutrient supply may affect microbially-mediated demethylation in wetlands and sediments. For example, soil saturation may promote the growth and metabolisms of anaerobic microorganisms, such as methanogens, which was considered as key players in MeHg degradation in alder swamps.[5, 21] Nutrient supply and abundance also affect bacterial activity, which in turn could either enhance or inhibit MeHg degradation.[22] For example, demethylation is found to increase at sites with elevated nutrient conditions.[23] However, despite the importance of the water and nutrients on soil microbial communities, their roles in regulating microbially-mediated MeHg degradation in paddy soils remain unexplored.

This study is therefore aimed at determining microbial communities associated with MeHg degradation and the influences of water saturation and nutrient addition (e.g., $NO_3^-$, $NH_4^+$, and Cu) on demethylation in paddy soils. Using both sterilized and unsterilized soils, we investigated the roles of microbes in MeHg degradation. We determined MeHg degradation kinetics as influenced by the addition of nutrients in the presence or absence of molybdate ($MoO_4^{2-}$) and bromoethanesulfonic acid (BES) as the respective inhibitors of sulfate-reducing bacteria and methanogens. Microbial community composition was identified using 16S rRNA Illumina sequencing and metatranscriptomic analyses, and multiple

92 statistical models were subsequently used to establish associations between microbial taxa
93 and MeHg degradation.

94 **MATERIALS AND METHODS**

95 **Experimental setup**

96 Soil samples (0~20 cm) were collected from selected paddy fields in Hunan (Taoyuan),
97 Guizhou (Wanshan), and Hubei (Enshi) provinces in Southwest China. These field sites are
98 more than 500 km apart from each other. The collected soil was sealed in plastic bags, placed
99 on ice, and then sent back to the laboratory for storage or processing. All samples were
100 homogenized, passed through 5.0 mm sieve to remove large plant residues and other
101 impurities, and finally divided into two sub-samples before use. One sub-sample was stored
102 at -20 °C for microbial analysis, and the other was stored at 4 °C for soil incubation
103 experiments. Additional site information and soil geochemical properties are provided in
104 Supporting Information Table S1.

105 We first investigated MeHg degradation in the three paddy soils via a short-term (48 h)
106 incubation experiment. The role of soil microbes in MeHg degradation was assayed using
107 both unsterilized and sterilized soils (treated by gamma rays with 0.6 kGy $hour^{-1}$ for 48
108 hours) prior to the addition of MeHg. Soil samples (20 g) from Hunan, Guizhou and Hubei
109 were respectively spiked with MeHg at the concentrations of 7.7, 6.8 and 5.6 $\mu g \cdot kg^{-1}$. The
110 spiked soil was then distributed to 100-mL serum bottles, which were flushed with pure
111 nitrogen gas for 10 min to produce an anoxic condition. Five replicates of each treatment
112 were incubated at 25 °C in the dark for 48 h. At the end of incubation, destructive sampling
113 was conducted to determine changes of soil MeHg concentrations.

114 The paddy soil from Hunan, representing one of the largest rice paddy soils in China,
115 was selected for a longer term (120 d) degradation experiment, aimed at identifying potential
116 microbial taxa related to MeHg degradation. The incubation experiments were conducted
117 with 3 controls and 8 treatments. The control samples included: (1) control, untreated soil
118 (CN, no MeHg added, at 70% water holding capacity); (2) control, water-unsaturated
119 condition (CT, at 70% water holding capacity); and (3) control, water-saturated condition
120 ($CT_W$). The soil was flooded with deionized water to maintain a standing water 5 cm level

above the soil surface.[24, 25] Here, we explored the effect of water condition on microbial degradation of MeHg since paddy soils are usually flooded in growing seasons but drained after rice harvest. The treated soils included: (4) the sterilized soil by gamma rays (Sterilization); (5) 40 mM sodium molybdate ($MoO_4$); and (6) 20 mM 2-bromoethylsulfonic (BES). Molybdate ($MoO_4^{2-}$) and BES were used to inhibit SRB and methanogens, respectively,[5, 10] as these microbes are known to influence both Hg methylation and MeHg degradation in soil. Additional treatments included: (7) addition of Cu at the concentration 400 mg·kg$^{-1}$ in soil to facilitate methanotrophs that could potentially demethylate MeHg;[11, 12] (8) 80 mM $NH_4Cl$ under water-saturated condition ($NH_{4W}$); (9) 200 mM $NaNO_3$ under water-saturated condition ($NO_{3W}$); (10) 80 mM $NH_4Cl$ under water-unsaturated condition ($NH_4$); and (11) 200 mM $NaNO_3$ under water-unsaturated condition ($NO_3$). $NH_4Cl$ and $NaNO_3$ were added to investigate the effect of N availability on microbial degradation of MeHg in the soil. MeHg was added into ten treatments soil at the concentration of 7.7 μg·MeHg kg$^{-1}$ soil, and CN treatment was not spiked with MeHg. Each treatment consisted of five replicates, each with 300 g soil that was evenly filled in the PVC (Polyvinylchloride) container with the dimension of 10 × 10 × 13 cm (length × width × height). All containers were soaked in 10% HCl solution for 24 h prior to use, and all treatments were incubated in the dark for 120 days at 25 °C. The water loss was measured by weighting the container, and replenished twice a week with deionized water. Soil samples were taken at 0, 2, 4, 7, 14, 30, 60, and 120 d, and used for measuring changes in MeHg concentration and soil chemical properties. Based on dynamic changes of MeHg concentrations, soils collected at 0, 60, and 120 d were selected for high-throughput 16S rRNA sequencing. In parallel, additional incubation experiments were performed to investigate expression changes of soil microbial community during MeHg degradation by metatranscriptomic analysis. Briefly, the Hunan soil (20 g) was homogenized and distributed to 100 mL anoxic serum bottles, with five replicates for each treatment, and incubated under the same conditions as used in the short-term experiment. Soil samples (5 g) were collected at 0 h ($T_0$) and 48 h ($T_{48}$) for total RNA extraction and metatranscriptomic analyses (Additional details provided in Supporting Information; Text S1).

**Soil geochemistry and MeHg analyses**

For MeHg analysis, 0.3~0.4 g soil was used to extract MeHg using $CuSO_4$-methanol/solvent method.[26, 27] An aliquot of 20~50 μL was then transferred to a 40 ml amber-glass vial, mixed with a citrate buffer and sodium tetraethylborate ($NaBEt_4$), and MeHg determined by an automated analyzer (Model III, Brooks Rand Labs). The method detection limit was 2 pg $L^{-1}$ for MeHg. Digestion blanks, reference material (ERM-CC580), and matrix spikes were analyzed along with samples for quality controls. The spiked recoveries of MeHg were between 94.5% and 101.9% (mean recovery: 98.2%). The analytical blank value for MeHg was < 2 pg $L^{-1}$. Total Hg (THg) was measured using a direct mercury analyzer (DMA-80, Milestone, Italy) with a detection limit of 0.1 ng $g^{-1}$.[28] Soil pH and Eh were monitored using an acidity meter (Sartorius PB-10, Germany). Ammonia and nitrate ($NH_4^+$ and $NO_3^-$) in soil were extracted with 2 M KCl at a solid-to-solution ratio of 1:5 by shaking at 180 rpm for 1 h and then measured by continuous flow analyzer (AA3, Germany SEAL).

**Soil DNA extraction and quantification of functional genes**

Total microbial DNA was extracted from 0.25 g of soil using the MoBio Power Soil DNA Isolation Kit (MoBio Laboratories, Carlsdad, CA, USA) according to the manufacturer's instructions. The concentration and quality of the DNA were checked using a NanoDrop ND-2000c UV-Vis spectrophotometer (NanoDrop Technologies, Wilmington, DE, USA). Key functional groups associated with Hg transformation (i.e. *dsrA*, *dsrB*, *mcrA*, *pmoA* and *hgcA*) were quantified in 25 μL PCR mixture using real-time quantitative PCR (qPCR, LightCycle 480 II, America), and the primer pairs and thermal cycling parameters are provided in Table S2. Standard curves were generated using the concentration and the threshold (Ct) and had an amplification efficiency ranging from 90% to 110%. The liner correlation coefficient ($R^2$) of all five genes ranged from 0.997~1.000.

**Soil microbial community analysis**

To characterize soil microbial community composition in different treatments, the V3-V4 region of the 16S rRNA gene was amplified using 338F/806R primers,[29] linking each barcode sequences (6 bp) to the forward primer for pooling multiple samples to one cube.

The PCR was performed using High-Fidelity PCR Master with a GC buffer. The reaction parameters were: 95 °C, 5 min; 94 °C, 30 s, 55 °C, 35 s, 72 °C, 30 s, 30 cycles; 72 °C, 8 min. After completion of the amplification, the target positive bands were quantified using Qubit, the same number of DNA were taken for mixing, and NEBNext Ultra assay was used to build the library. Qualified libraries were sequenced using Illumina HiSeq 2500 (PE250). The raw data obtained by sequencing was subjected to quality control by ngs toolkit software to remove low-quality sequences,[30] then spliced by Flash software,[31] and the data was processed using Qiime.[32] The resulting clean tags were cleaned with Mothur software (version v.1.30.1, USA),[33] the forward primer and reverse primer sequences were removed, the data set was annotated using the Silva (db128) ribosomal RNA gene database project,[34] and the OTU was partitioned at 97% using the vsearch program.[35]

**Statistical analysis**

ANOVA analysis was used to test for significant differences between control and treatments. Statistical tests were carried out at the 0.05 significance level in general, and the significance level of 0.01 was referred to as highly significant. A classification random forest (RF) analysis was used to identify microbial taxa (bacteria families) associated with MeHg degradation. The analysis aimed to test which microbial taxa are possible predictors of MeHg degradation. Compared to the percentage increases in the mean squared error (MSE) of MeHg degradation to estimate the importance of the microbial taxa: the higher MSE% values, the more important variables.[36] We also identified the taxa that potentially associated with MeHg degradation through the analysis of microbial network. We first selected the OTUs with a relative abundance of more than 0.1%, then calculated all pairwise Spearman's rank correlations between all soil bacterial taxa using the psych package of the R statistical software. A co-occurrence is considered to be stable if the Spearman's correlation value > 0.5 and $p < 0.01$. The network was visualized with the interactive platform Gephi, and used default parameters from the interactive platform Gephi to identify ecological clusters (i.e., modules) of soil taxa strongly interacting with each other. The relative abundance of each module was computed by averaging the standardized relative abundances (z-score) of the OTUs affiliated with each ecological cluster.[37] There was a significant positive correlation

between the relative abundance of the Mod 3 and the MeHg degradation efficiency. We then conducted a random forest analysis to identify potential taxa related to MeHg degradation in Mod 3. Linear discriminant analysis (LDA) effect size (LEfSe) analysis was used to identify bacterial biomarkers for 2 groups on the open website (http://huttenhower.sph.harvard.edu/galaxy). We set an α-value < 0.05, and the threshold used to consider a discriminative feature for the logarithmic LDA score was set at > 2. R package DESeq2 was performed to identify the differentially expressed genes between two time points ($T_0$ and $T_{48}$).

## RESULTS AND DISCUSSION

### MeHg degradation and microbial community analysis in soil

Results showed that MeHg concentrations sharply decreased at 48 h in all unsterilized soils from Hunan, Guizhou and Hubei provinces (Figure S1), whereas no significant changes ($p > 0.05$) were observed for the sterilized soils. Negligible differences were observed in MeHg degradation efficiencies among the paddy soils from distinct geographic sites, and they ranged from 30.5 to 33.9%. These results indicate that soil microbes play a key role in the degradation of MeHg, which is consistent with pure culture studies showing that microbes can degrade MeHg in a short time.[7, 11]

To explore possible mechanisms and microbes involved in MeHg degradation, longer incubation experiments were subsequently performed using the Hunan paddy soil. The MeHg content in the sterilized soil did not significantly change (~ 9%, $p > 0.05$) from day 0 to 60 despite some variations observed in the first week (Figure 1a). In addition, soil MeHg concentrations in both $CT_W$ (water-saturated) and CT (water-unsaturated) treatments decreased consistently with time. At day 60, more than 72.71% of MeHg in the $CT_W$ was degraded, which was much higher ($p < 0.01$) than that in the CT (55.44%), suggesting water saturation enhanced MeHg degradation in the soil. MeHg contents continued to decrease over time (up to 120 d) in the $CT_W$ (79.63%) and CT (74.78%) treatments. Interestingly, MeHg concentration in the sterilized treatment also decreased at day 120, albeit lower than those observed in the unsterilized treatments. This result could be attributed to a resilience of microbial community over a long incubation time despite initial suppression, as previously

reported.[38, 39] Thus, we further conducted microbial analysis for the sterilized soil samples at 120 d. Results of qPCR showed the presence of Hg-transformation related genes (*dsrA*, *dsrB*, *mcrA*, *pmoA*) in the soil (Figure S2a), although their abundances were much lower than those at day 0. Also, we detected diverse microbial taxa in the sterilized soil (Figure S2b), and some of them are key members that are known to carry *merA* genes (i.e., *Alphaproteobacteria*, *Actinobacteria*, *Gammaproteobacteria*, *Betaproteobacteria* and *Bacteroidetes*).[16] Together, these results suggest that microbial community recovered to some extent in the sterilized soil after 120 d, which could explain the decrease of MeHg concentrations. These observations are consistent with the analyses of the abundances of Hg transformation genes, which were much higher in the unsterilized soil than those in the sterilized soil ($p < 0.05$) (Figure S3). Soil sterilization significantly reduced the abundance of SRB (*dsrA* and *dsrB*), methanogens (*mcrA*) and methanotrophs (*pmoA*), which have all been implicated for MeHg degradation,[10, 11] though the underlying mechanisms remain unclear. These results further demonstrate important roles of microbial community in the degradation of MeHg in paddy soils, agreeing with previous studies highlighting that soil microbes could be crucial for MeHg degradation in the environment.[5, 6, 9, 40]

To explore the microbial taxa associated with MeHg degradation, we characterized changes in soil bacterial communities at days 60 and 120. The LEfSe analysis was used to assess which microbiota changed during incubation between the initial soil (i.e. CN) and the water-saturated soil ($CT_W$, MeHg-spiked) (Figure 1b). Results showed that *Saccharibacteria*, *Acidobacteria*, *Actinobacteria* and *Proteobacteria* were the major biomarkers in the CN treatment. Much more biomarkers, including *Bacteroidetes*, *Ignavibacteria*, *Nitrospirae*, *Firmicutes*, etc., were observed in the $CT_W$ treatment in addition to those found in the CN treatment. Among the biomarkers in the $CT_W$ treatment, *Bacteroidetes* and *Firmicutes* were identified as potential members containing the *merB* gene that can cleave MeHg, although we were unable to identify the biomarkers at higher taxonomic levels due to limited information on *merB*. Compared to the CT, the LEfSe analysis revealed that *Gracilibacteria* was the biomarker for the CN group at 120 d, and the biomarker in the CT group was *Armatimonadetes* (Figure S4). The abundances of dominant phyla (i.e. beta-/delta-*Proteobacteria*, *Bacteroidetes*) in the $CT_W$ soil were also higher than those in the

CT soil (Figure 1c), agreeing with the biomarkers of LEfSe analysis between $CT_W$ and CT (Figure 1d). These results also agree with the observation that more MeHg was degraded in the $CT_W$ than in the CT soil after 4 days (Figure 1a), suggesting that anoxic water-logged conditions likely favored anaerobic microbes in the degradation of MeHg under anoxic conditions. On the other hand, anoxic conditions also benefit Hg methylation by anaerobic microorganisms,[7, 8] and the two processes usually happen simultaneously in natural environments. However, contributions of Hg methylation under these conditions should have little effects on our observations due to a low level of available Hg in the soil.[41, 42]

**Effects of Cu, N, $MoO_4$, and BES on MeHg degradation and microbial communities**

Under unsaturated conditions, methanotrophs have been implicated in degrading MeHg, which is regulated by the availability of Cu.[11, 12] Our results show that the addition of Cu promoted degradation of MeHg ($p < 0.05$). There was a sharp decrease (76%) in MeHg within 2 days, after that MeHg concentrations varied slightly between 2 and 120 days (Figure 2a). Microbial communities in the control soil (CT) were dominated by *Proteobacteria*, *Acidobacteria*, *Actinobacteria*, *Chloroflexi*, *Firmicutes*, *Bacteroidetes*, while addition of Cu remarkably increased the relative abundances of gamma-*Proteobacteria*, *Firmicutes* and *Actinobacteria* (Figure 2b; Figure S5a). In particular, treatment with Cu strikingly enriched the family *Xanthomonadaceae* (gamma-*Proteobacteria*, Figure 2c). These findings agree with a recent study showing an association between *Xanthomonadaceae* (a member of *Xanthomonadales*) and MeHg concentrations in the paddy soil,[43] although its roles in influencing MeHg degradation and the underlying mechanisms require further investigation. The *Xanthomonadaceae* consists mainly of *Dyella* genus, which was not detected in other treatments (Figure S5b). Previous studies suggested that *Dyella* spp. could degrade biphenyl in sludge samples via dioxygenation at the 2,3-position to gain carbon and energy.[44, 45] Genome analysis of the *Dyella* revealed the presence of a series of copper-resistant cus genes in the taxa,[46] which may partially explain why *Dyella* was significantly enriched at the Cu treated soil. These observations demonstrate the importance of soil gamma-*Proteobacteria* in the degradation of MeHg, although whether or not Cu facilitated the microbial demethylation via methanotrophs requires further investigation as we did not observe an increase of *pmoA*

abundance with the Cu addition (Figure S6a). So far, only certain groups of methanotrophs are shown to degrade MeHg,[11] and we cannot identify these methanotrophs affected by Cu addition due to lack of specific-marker genes. Furthermore, most of the examined genera are close to type I methanotrophs (Table S3), which do not have the ability to degrade MeHg.[11] Our results therefore suggest that Cu may have promoted MeHg degradation via other unknown communities.

Under water-saturated conditions, soil MeHg concentrations decreased with extended incubation time, regardless of treatments with $MoO_4$, BES, $NH_4^+$, or $NO_3^-$ (Figure 2d). The microbial community composition (e.g. percentage of bacteria phylum and genus) did not change in the $MoO_4$ and BES treatments (Figure S7), although the addition of $MoO_4$ and BES significantly inhibited SRB and methanogens, respectively (i.e., copy numbers of *dsrB* and *mcrA*) (Figure S6b, c). These results suggest that SRB or methanogens alone could not account for MeHg degradation in the paddy soil. These observations are different from earlier studies suggesting that SRB and methanogens play an important role in the degradation of MeHg in swamp and lake sediments.[5,9,10] SRB and methanogens degrade MeHg mainly via the oxidative pathway in the environment at relatively low Hg concentrations,[14] and thus may not be the key microbial taxa responsible for MeHg degradation in the paddy soils contaminated with high concentrations of MeHg. We did not observe significant effects of N addition on that microbial MeHg degradation in the unsaturated and saturated soils (Figure 2a, d), although N fertilizers are intensively applied to paddy fields for rice productivity.[47, 48] and are considered to be a key factor regulating soil microbial community.[18, 19] The result is in contrast with an earlier study where demethylation rate is found to increase with increased nutrient conditions (e.g. C/N) in Alder swamps.[5] A possible explanation could be that the soil used in the present study was from farmlands and already contained an excess amount of nutrients, and thus the addition of N did not further promote the activity of the microbes associated with MeHg degradation. This argument is also supported by no significant changes in the abundances of Hg-transformation associated microbial groups and the composition of bacteria communities (Figure S8, S9). Previous studies also suggest that N ($NH_4^+$ or $NO_3^-$) addition could influence soil methanotrophic communities,[49, 50] but no significant influences were observed either in the present study.

## Microbial taxa associated with MeHg degradation

We subsequently used RF statistical analysis to investigate which microbial taxa (bacteria families) may be associated with MeHg degradation. Results showed that *Proteobacteria*, *Actinobacteria,* and *Firmicutes* were mainly associated with MeHg degradation (Figure S10). Due to lack of marker genes of microbial demethylation, we annotated potential taxa associated with MeHg degradation by grouping bacterial OTUs into different ecological clusters (or modules, Mod). Each Mod comprised of strongly co-occurring bacterial taxa with one another (Figure 3a), and they could share common ecological functions such as the degradation of organic compounds.[51] The results indicated that MeHg degradation was positively correlated with the relative abundance of microbial taxa belong to Mod 3 ($p < 0.05$) (Table S4). Additional RF analysis showed that MeHg degradation efficiency was associated with 19 genera from Mod 3, and the *Clostridium_sensu_stricto_1* had the strongest correlation with MeHg degradation (Figure 3b and Table S5). We also found that these genera were mainly affiliated with *Actinobacteria*, and similar results were observed when carrying out a run of RF analysis at the family level (Figure S11). As an important member carrying MerA,[16] *Actinobacteria* may thus be the key phylum associated with reductive demethylation in the paddy soil. In addition, *Actinobacteria* has been widely reported to be able to degrade organic matter (i.e. cellulose),[52, 53] and may thus have the potential in degrading organic compounds such as MeHg.

To provide additional insights into active microbial groups associated with MeHg degradation, RNA-based metatranscriptomic analysis was performed. The LEfSe analyses identified 4 biomarkers for $T_0$ group and 12 biomarkers in the $T_{48}$ group at the phylum level (Figure S12). To investigate the specific microbial communities associated with MeHg degradation, additional LEfSe analysis was performed at the family level. Ten biomarkers were observed in the $T_0$ and $T_{48}$ group, respectively (Figure 4a). Among the biomarkers of $T_{48}$ group, *Catenulisporaceae*, *Frankiaceae*, *Mycobacteriaceae* and *Thermomonosporaceae* have been identified as the most probable predictors of MeHg degradation based on Illumina sequencing analysis of 16S rRNA. These results cross-validated our findings that these microbial taxa were most likely implicated in MeHg degradation in the paddy soil. However,

we note that the abundances of these taxa are relatively low in the soil (Table S6), and there could be some uncharacterized taxa involved in MeHg degradation, which require additional investigation. Different biomarkers of MeHg degradation may exist in different types of paddy soils, as previous studies have revealed the existence of different microbial communities in paddy soils from different geographical locations.[54, 55] For example, relative abundances of *Catenulisporaceae* and *Frankiaceae* in the soils of Guizhou (0.002% and 0.03%, respectively) and Hubei (< 0.0001% and 0.005%, respectively) are found to be much lower than those in the Hunan paddy soil.[27, 43] It suggests that some unknown microbial taxa may also be involved in MeHg degradation in the different types of paddy soils. Additional studies are obviously warranted to further explore microbial communities associated with MeHg degradation and the underlying mechanisms in a variety of paddy soils.

In addition, we found that 1017 genes were up-regulated and 28 genes were down-regulated in the $T_{48}$ treatment (Figure 4b and Table S7). Some of the up-regulated genes are involved in microbial metabolisms (e.g., nitrotoluene degradation, fatty acid degradation), genetic information processing (e.g., sulfur relay system, ribosome) and environmental information processing (e.g., ABC transporters). The down-regulated genes were mainly implicated in amino acid metabolism. In particular, the 2-oxoglutarate ferredoxin oxidoreductase (encoded by *afw* gene) was significantly up-regulated, which may be associated with Hg transformation according to a recent study, which showed that MeHg formation might be relevant to iron transport system.[56] We also using hidden Markov models (HMMs) to identify *merA* and *merB* homologs from metatranscriptomic data. The results showed hundreds of *merA* homologs, but *merB* homologs were rarely detected in the soil (Table S8). However, there were no significant differences in the expression of *merA* and *merB* between $T_0$ and $T_{48}$, suggesting that reductive demethylation was not a dominant pathway of MeHg degradation in the soil, at least in short-term (48 h) incubations. Additionally, we did not observe significant changes of the THg concentration during 120 d incubation (Figure S13), despite a high abundance of *merA* was found by the analysis of metatranscriptomic data.

## ENVIRONMENTAL IMPLICATIONS

This study demonstrates important roles of microbial communities in MeHg degradation in paddy soils, similarly as observed in wetlands and some pure culture studies.[5, 7] Illumina sequencing and metatranscriptomic analyses consistently identified some microbial taxa associated with MeHg degradation such as *Catenulisporaceae*, *Frankiaceae*, *Mycobacteriaceae*, and *Thermomonosporacea*, which may be used as potential predictors of MeHg degradation in paddy soils. These findings also have implications for other MeHg contaminated habitats such as wetlands and sediments, owing to the ubiquity of the microbial community in the ecosystems. However, future studies are needed to validate whether or not these identified microbial taxa are involved in degrading MeHg directly or through indirect manners.[43] Our results could therefore serve as a guidance in searching for MeHg degrading bacteria, thereby advancing our knowledge on demethylation biomarkers in paddy soils.

Microbial communities in paddy soils are mainly regulated by water and nutrient conditions.[19, 20] Our results suggest that water saturation facilitated microbially-mediated MeHg degradation in paddy soils. This result could provide new insights into microbial degradation of MeHg in terrestrial ecosystems, as well as the management of Hg contaminated farmlands, although water saturation also favors MeHg biosynthesis by anaerobic microbes.[8, 57] Paddy field is a typical agricultural ecosystem which has a relatively high level of nutrients, but our results suggest that addition of N had no effects on microbial taxa associated with MeHg degradation in soil. Therefore, these observations further emphasize the importance of water management for mitigation of MeHg in soil. On the other hand, Cu is found to enhance MeHg degradation by soil microorganisms, suggesting the importance of environment factors in influencing microbial communities involved in MeHg degradation, particularly at Hg contaminated sites with high level of metal ions, such as Cu.[56] Overall, we provide the first comprehensive characterization of microbial communities and environmental factors potentially associated with MeHg degradation in paddy soils. These results have important implications in developing strategies of MeHg mitigation in paddy soils, such as improved management of water saturation, fertilization, and Cu addition, thereby reducing MeHg exposure to billions of people through rice consumption.

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## ASSOCIATED CONTENT

### Supporting Information and Data Availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Materials. This material is available free of charge via the Internet at http://pubs.acs.org.

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**FIGURE LEGENDS:**

**Figure 1.** (a) The effects of water management (unsaturated and saturated condition) and sterilization on the dynamics of MeHg degradation; (b) LEfSe (linear discriminative analysis effect size) analysis for taxonomic biomarkers between no MeHg addition and the water-saturated treatments at the phylum level; (c) the dominant bacteria phyla in the water-saturated and unsaturated treatments; (d) LEfSe analysis for taxonomic biomarkers on phylum level between the water-unsaturated and water-saturated treatments. CN represents the treatment with no MeHg addition, CT represents water-unsaturated treatment, $CT_W$ represents water-saturated treatment, and Sterilization represents sterilized treatment.

**Figure 2.** (a) The effects of Cu and nitrogen ($NH_4^+$ and $NO_3^-$) addition on the dynamics of MeHg degradation in the water-unsaturated treatment; (b-c) The dominant bacteria phyla and families at Cu addition treatment, respectively; (d) The effects of inhibitors ($MoO_4$ and BES) and nitrogen ($NH_4^+$ and $NO_3^-$) addition on the dynamic of MeHg degradation in the water-saturated treatment (i.e. $NH_{4W}$ an $NO_{3W}$, respectively).

**Figure 3.** (a) A microbial network diagram with nodes (taxa) colored by each of the major four ecological clusters (modules, Mod) of bacterial communities; (b) Random forest analysis identifying potential genera of Mod 3 (top 10) associated with MeHg degradation.

**Figure 4.** (a) The LEfSe analysis for taxonomic biomarkers at the family level and (b) top 20 differently expressed genes with R package DESeq2 at the $T_{48}$ treatment. The identified taxa were derived from RNA-based metatranscriptomic analysis.

Figure 1

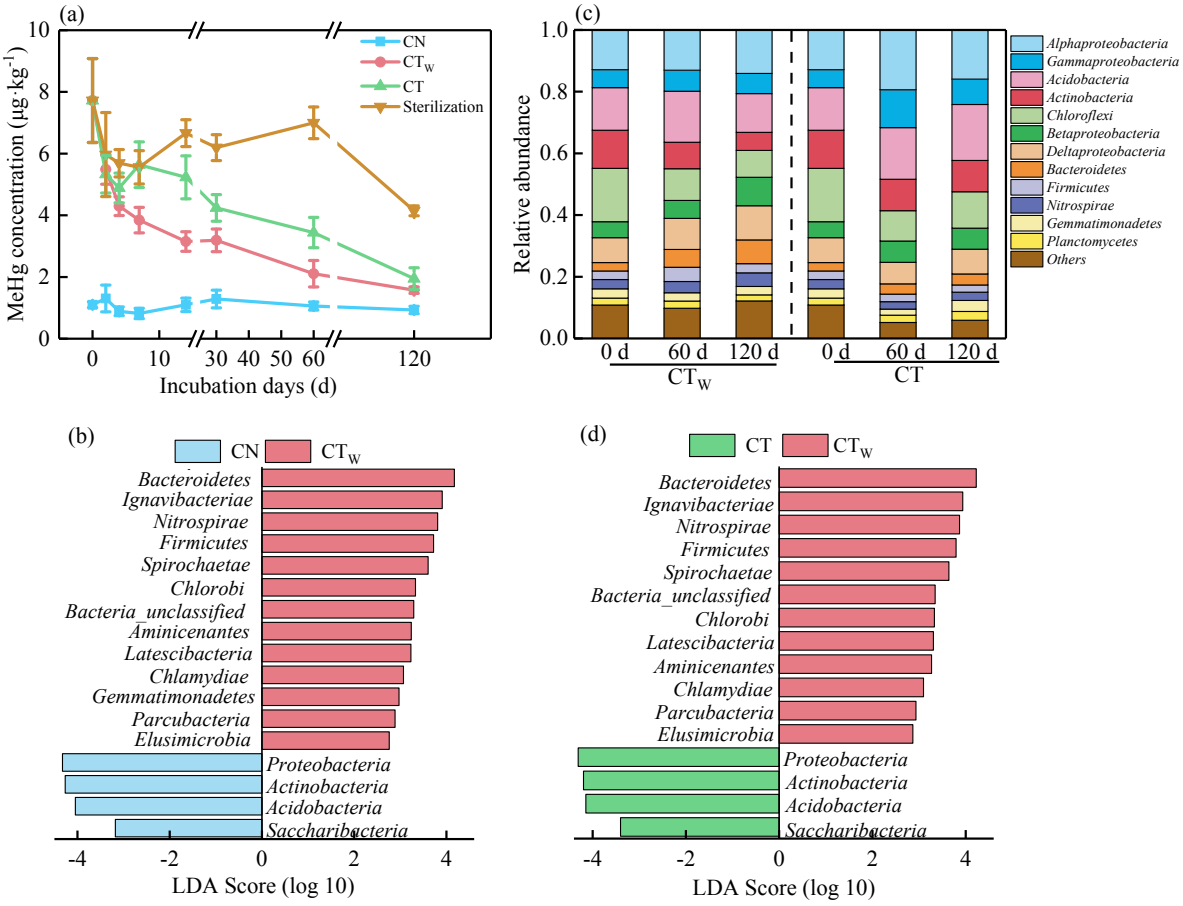

(a)
MeHg concentration (μg·kg-1)
Incubation days (d)
CN
CTW
CT
Sterilization
(b)
CN
CTW
Bacteroidetes
Ignavibacteriae
Nitrospirae
Firmicutes
Spirochaetae
Chlorobi
Bacteria_unclassified
Aminicenantes
Latescibacteria
Chlamydiae
Gemmatimonadetes
Parcubacteria
Elusimicrobia
Proteobacteria
Actinobacteria
Acidobacteria
Saccharibacteria
LDA Score (log 10)
(c)
Relative abundance
0 d
60 d
120 d
CTW
CT
Alphaproteobacteria
Gammaproteobacteria
Acidobacteria
Actinobacteria
Chloroflexi
Betaproteobacteria
Deltaproteobacteria
Bacteroidetes
Firmicutes
Nitrospirae
Gemmatimonadetes
Planctomycetes
Others
(d)
CT
CTW
LDA Score (log 10)

Figure 2

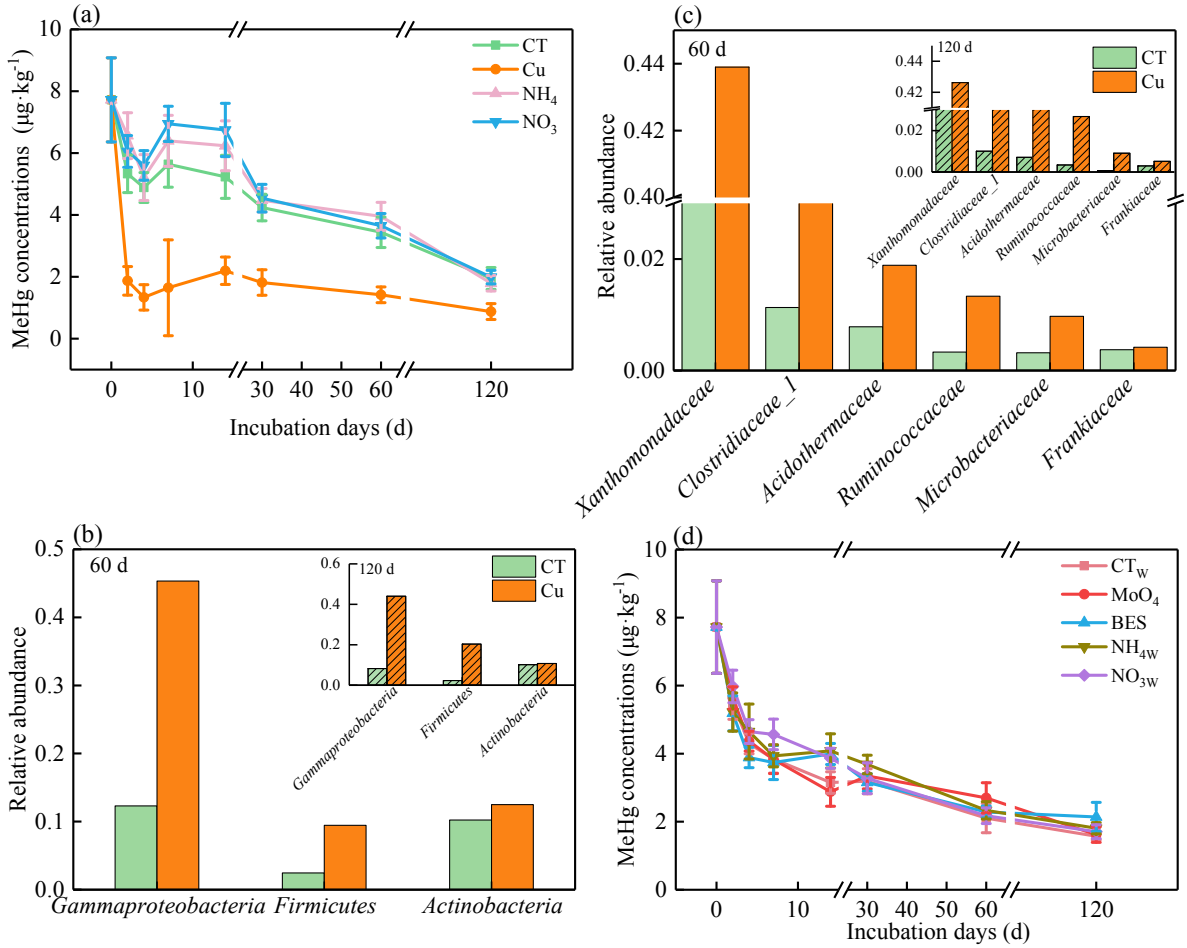

(a)
MeHg concentrations (μg·kg-1)
Incubation days (d)
CT
Cu
NH4
NO3
(c)
60 d
120 d
Relative abundance
Xanthomonadaceae
Clostridiaceae_1
Acidothermaceae
Ruminococcaceae
Microbacteriaceae
Frankiaceae
(b)
Gammaproteobacteria
Firmicutes
Actinobacteria
(d)
CTW
MoO4
BES
NH4W
NO3W

Figure 3

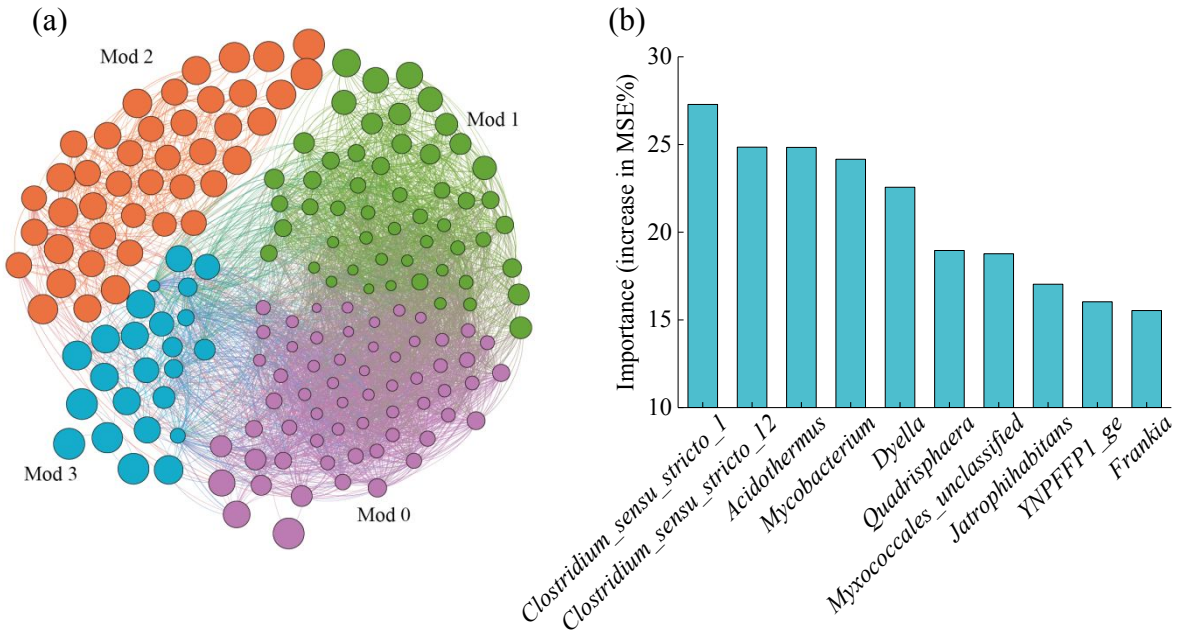
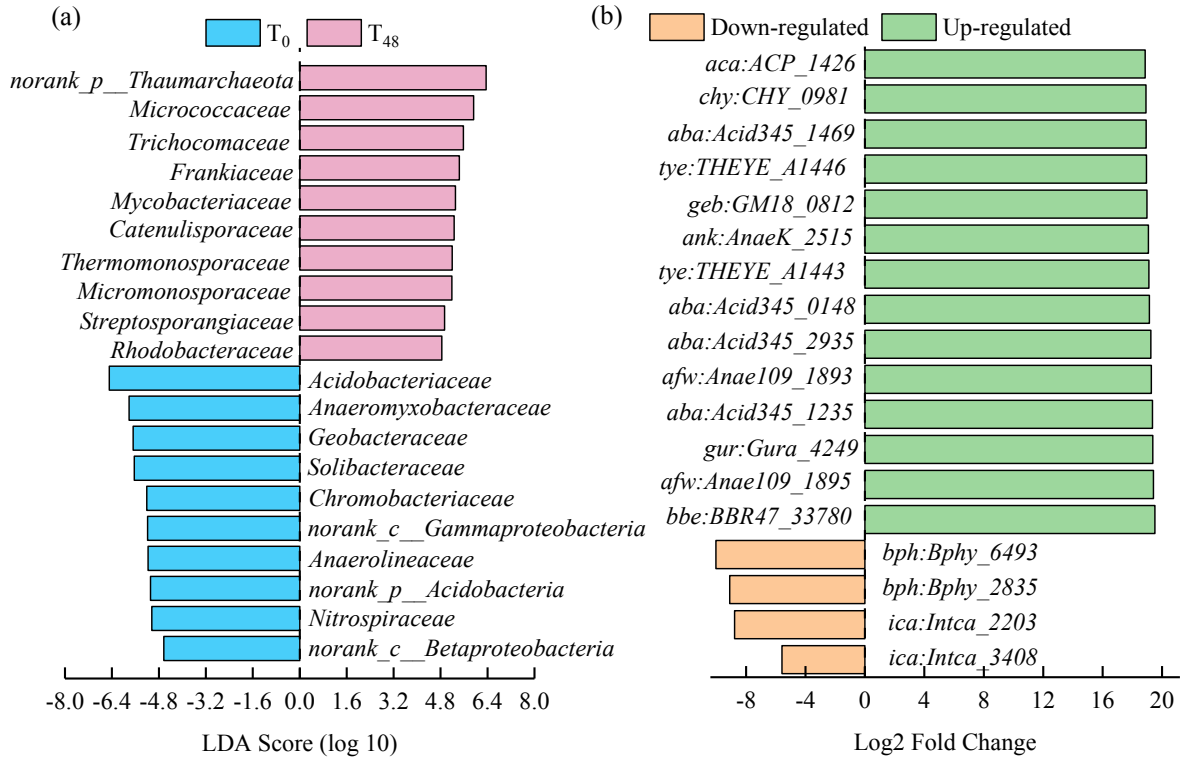
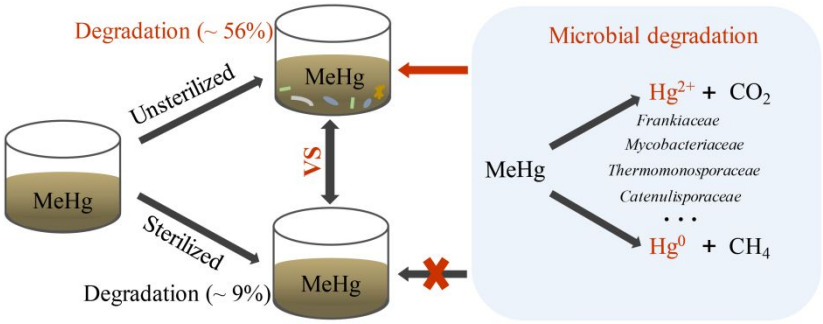

(a)
Mod 2
Mod 1
Mod 3
Mod 0
(b)
Importance (increase in MSE%)
30
25
20
15
10
Clostridium_sensu_stricto_1
Clostridium_sensu_stricto_12
Acidothermus
Mycobacterium
Dyella
Quadrisphaera
Myxococcales_unclassified
Jatrophihabitans
YNPFFP1_ge
Frankia

Figure 4


(a)
T0
T48
norank_p__Thaumarchaeota
Micrococcaceae
Trichocomaceae
Frankiaceae
Mycobacteriaceae
Catenulisporaceae
Thermomonosporaceae
Micromonosporaceae
Streptosporangiaceae
Rhodobacteraceae
Acidobacteriaceae
Anaeromyxobacteraceae
Geobacteraceae
Solibacteraceae
Chromobacteriaceae
norank_c__Gammaproteobacteria
Anaerolineaceae
norank_p__Acidobacteria
Nitrospiraceae
norank_c__Betaproteobacteria
-8.0 -6.4 -4.8 -3.2 -1.6 0.0 1.6 3.2 4.8 6.4 8.0
LDA Score (log 10)
(b)
Down-regulated
Up-regulated
aca:ACP_1426
chy:CHY_0981
aba:Acid345_1469
tye:THEYE_A1446
geb:GM18_0812
ank:AnaeK_2515
tye:THEYE_A1443
aba:Acid345_0148
aba:Acid345_2935
afw:Anae109_1893
aba:Acid345_1235
gur:Gura_4249
afw:Anae109_1895
bbe:BBR47_33780
bph:Bphy_6493
bph:Bphy_2835
ica:Intca_2203
ica:Intca_3408
-8 -4 0 4 8 12 16 20
Log2 Fold Change


Degradation (~ 56%)
MeHg
Unsterilized
VS
MeHg
Sterilized
Degradation (~ 9%)
MeHg
Microbial degradation
MeHg
$Hg^{2+}$ + $CO_2$
Frankiaceae
Mycobacteriaceae
Thermomonosporaceae
Catenulisporaceae
• • •
$Hg^0$ + $CH_4$

TOC Art